Mohammad Mahmudul Hasan, PhD

Associate Professor
Department of Electrical and Electronic Engineering

Ibsens gate 13A, Gjøvik 2821, Norway
(+47) 46 34 56 32 / (+880) 1710 037000
mohammad.m.hasan@ntnu.no
NTNU · UITS · Portfolio
Google Scholar · LinkedIn

Education

Norwegian University of Science and Technology, Norway **(05.2022 – 06.2025)**
Doctor of Philosophy
Information and Communication Technology

KIIT University, India **(07.2008 – 06.2010)**
Master of Technology
Communication Systems Engineering

KIIT University, India **(07.2004 – 06.2008)**
Bachelor of Technology
Electronics and Telecommunication Engineering

Work Experience

University of Information Technology and Science, Bangladesh **(01.2011 – Present)**
- *Associate Professor*, Electrical and Electronic Engineering — 07.2025 – Present
- *Assistant Professor*, Electrical and Electronic Engineering — 01.2011 – 06.2025

Norwegian University of Science and Technology, Norway **(05.2022 – 06.2025)**
- *Research Fellow*, Faculty of Engineering

Brno University of Technology, Czech Republic **(08.2024 – 01.2025)**
- *Visiting Research Fellow*, Department of Automation

KIIT University, India **(02.2009 – 12.2010)**
- *Assistant Professor*, Electronics and Telecommunication Engineering — 07.2010 – 12.2010
- *Teaching Assistant*, Electronics and Telecommunication Engineering — 02.2009 – 06.2010

Selected Publications

- Inaamullah Khan, **Mohammad Mahmudul Hasan**, Michael Cheffena, "Transmitter-Assisted Joint Data-Aided Channel Estimation and PAPR Reduction Scheme in Wireless Fading Channels", **Scientific Reports**, vol. 16, no. 1, 8015, 2026 **(2026)**
- **Mohammad Mahmudul Hasan**, Onur Alev, Pavel Skrabanek, Gabriela Soukupová, Fatima Hassouna, and Michael Cheffena, "Microwave MIMO E-Nose for Wireless Communication and Selective Detection of VOC Mixtures with Concentration Estimation", **ACS Sensors**, vol. 10. no. 9, pp. 6446–6463, 2025 **(2025)**
- Inaamullah Khan, **Mohammad Mahmudul Hasan**, Michael Cheffena, "A Novel Low-Complexity Peak-Power-Assisted Data-Aided Channel Estimation Scheme for MIMO-OFDM Wireless Systems", **IEEE Open Journal of Signal Processing**, vol. 6, pp. 992 – 1003, 2025 **(2025)**
- Onur Alev, **Mohammad Mahmudul Hasan**, Emel Tuğba Ertuğrul, Selçuk Birdoğan, Okan Özdemir, Eda Goldenberg, Michael Cheffena, "Hydrothermally Synthesized Molybdenum disulfide Nanoflakes: Structural, Electrical, and Antenna-based Gas Sensing Characteristics", **Sensors & Actuators: A. Physical**, vol. 393, 116756, 2025 **(2025)**
- **Mohammad Mahmudul Hasan**, Onur Alev, Pavel Skrabanek, and Michael Cheffena, "Molecularly Imprinted Polymer-Based Electronic Nose for Ultrasensitive, Selective Detection and Concentration Estimation of VOC Mixtures", **IEEE Sensors Journal**, vol. 25, no. 10, 2025 **(2025)**
- **Mohammad Mahmudul Hasan**, Todd Cowen, Onur Alev and Michael Cheffena, "MIMO Microwave Sensor for Selective and Simultaneous Detection of Methanol and Ethanol Gases at Room Temperature," **IEEE Transactions on Instrumentation & Measurement**, vol. 74, 9511613, 2025 **(2025)**
- **Mohammad Mahmudul Hasan**, Onur Alev and Michael Cheffena, "Dual-Functional Antenna Sensor for Highly Sensitive and Selective Detection of Isopropanol Gas Using Optimized Molecularly Imprinted Polymers," **ACS Sensors**, vol. 10, no. 3, pp. 2147–2161, 2025 **(2025)**

- **Mohammad Mahmudul Hasan**, Onur Alev, Eda Goldenberg and Michael Cheffena, "MoS2/MoOx Nanoflake-Based Dual-Functional Antenna Sensors for Highly Sensitive and Selective Detection of Volatile Organic Compounds," **ACS Applied Nano Materials**, vol. 7, no. 21, pp. 25065–25077, 2024 **(2024)**
- **Mohammad Mahmudul Hasan**, Todd Cowen and Michael Cheffena, "A Novel Molecularly Imprinted Polymer-Based Carbon Nanotube-Coated Microwave Sensor for Selective Detection of Methanol Gas," **IEEE Sensors Letters**, vol. 8, no. 5, 6004904, 2024 **(2024)**
- **Mohammad Mahmudul Hasan**, Onur Alev, Eda Goldenberg and Michael Cheffena, "A Novel Molybdenum Disulfide-Based High-Precision Microwave Sensor for Methanol Gas Detection at Room Temperature," **IEEE Microwave and Wireless Technology Letters**, vol. 34, no. 6, pp. 691 – 694, 2024 **(2024)**
- **Mohammad Mahmudul Hasan**, Michael Cheffena, "Adaptive Antenna Impedance Matching Using Low-Complexity Shallow Learning Model", **IEEE Access**, vol. 11, pp. 74101 – 74111, 2023 **(2023)**
- **Mohammad Mahmudul Hasan**, Michael Cheffena, Slobodan Petrovic, "Physical-layer Security Improvement in MIMO OFDM Systems using Multilevel Chaotic Encryption", **IEEE Access**, vol. 11, pp. 64468 – 64475, 2023 **(2023)**
- Inaamullah Khan, Michael Cheffena, **Mohammad Mahmudul Hasan**, "Data Aided Channel Estimation for MIMO-OFDM Wireless Systems Using Reliable Carriers", **IEEE Access**, vol. 11, pp. 47836 – 47847, 2023 **(2023)**
- **Mohammad Mahmudul Hasan**, Mohammad Mahadi Hasan Foad, "Modified Gamma Correction Companding for PAPR Reduction in OFDM Systems Considering Solid State Power Amplifier and Wireless Channels", **Circuits, Systems, and Signal Processing**, vol. 37, no. 10, pp. 4431- 4454, 2018 **(2018)**
- **Mohammad Mahmudul Hasan**, Mohammad Mahdi Hasan Faisal, "IGCC for PAPR Reduction in OFDM Systems over the Nonlinearity of SSPA and Wireless Fading Channels", **Circuits, Systems, and Signal Processing**, vol. 35, no. 8, pp. 2855–2880, 2015 **(2015)**
- **Mohammad Mahmudul Hasan**, "PAR Reduction in SU/MU-MIMO OFDM Systems using OPF Precoding over the Nonlinearity of SSPA", **Wireless Personal Communications**, vol. 83, no. 3, pp. 2225-2248, 2015 **(2015)**
- **Mohammad Mahmudul Hasan**, "A Novel CVM Precoding Scheme for PAPR Reduction in OFDM Transmissions", **Wireless Network**, vol. 20, no. 6, pp. 1573-1581, 2014 **(2014)**
- **Mohammad Mahmudul Hasan**, "A New PAPR Reduction Scheme for OFDM Systems Based on Gamma Correction", **Circuits, Systems, and Signal Processing**, vol. 33, no. 5, pp. 1655-1668, 2014 **(2014)**
- **Mohammad Mahmudul Hasan**, "A New PAPR Reduction Technique in OFDM Systems Using Linear Predictive Coding", **Wireless Personal Communications**, vol. 75, no. 1, pp. 707-721, 2014 **(2014)**
- **Mohammad Mahmudul Hasan**, "VLM Precoded SLM Technique for PAPR Reduction in OFDM Systems", **Wireless Personal Communications**, vol. 73, no. 3, pp. 791-801, 2013 **(2013)**
- **Mohammad Mahmudul Hasan**, "PAPR Reduction in OFDM Systems Based on Autoregressive Filtering", **Circuits, Systems, and Signal Processing**, vol. 33, no. 5, pp. 1637-1654, 2013 **(2013)**

See more at **Google Scholar**

Research Projects and Initiatives

- **Machine Learning Assisted Signal Processing for Antenna-Sensor Arrays** **(2024 – 2025)**
  Work Package Member, The Research Council of Norway (353890)
- **Air Quality Monitoring Exploiting Massive Multiple-Input Multiple-Output Antennas** **(2022 – 2025)**
  Work Package Member, The Research Council of Norway (324061)
- **University Comprehensive Academic Management- Software Implementation Project** **(2020 – 2022)**
  Project Manager, University of Information Technology and Sciences (UITS)
- **Higher Education Quality Enhancement Project (HEQEP)** **(2017 – 2018)**
  Member, University Grants Commission, Bangladesh and the World Bank

Academic Leadership and Professional Service

University of Information Technology and Science, Bangladesh **(01.2011 – Present)**
- *Advisor*, Information and Communication Technology Cell — 01.2018 – Present
- *Head of the Department*, Electronics and Communication Engineering — 02.2019 – 04.2022
- *Director*, Information and Communication Technology Cell — 12.2020 – 04.2022
- *Editorial Board Member*, Journal of Science and Engineering — 10.2020 – 04.2022
- *Member*, Internal Quality Assurance Cell (IQAC) — 07.2017 – 07.2018
- *Member*, Finance Committee — 03.2020 – 04.2022
- *Member*, Academic Council — 02.2019 – 04.2022

Sanyo Engineering & Construction Inc., Japan **(04.2018 – 01.2019)**
- *Consultant and Technical Trainer*, BJIT Limited, Bangladesh

Professional Certifications

Brno University of Technology, Czech Republic **(08.2024 – 01.2025)**
Professional Competence, Electrical Engineering – NV No. 194/2022 Coll.

GrameenPhone Ltd., Bangladesh **(07.2007 – 10.2007)**
Intern Engineer, Transmission Planning Division

All India Radio & Doordarshan (Prasar Bharati), India **(05.2007 – 07.2007)**
Industrial Training, Broadcasting Corporation of India

Red Hat Bhubaneswar, India **(04.2006 – 07.2006)**
Industrial Training, Red Hat Linux RHEL 4

Grants & Awards

Research Recognition **(06.2025)**
Nominated for *Best Doctoral Thesis* in Sensors
Norwegian University of Science and Technology

Research Grant **(08.2024)**
Awarded 146 000 NOK
The Research Council of Norway

Chancellor's Gold Medal **(12.2010)**
Awarded for *securing the highest CGPA* (10/10)
KIIT University, India

Founder's Gold Medal **(12.2010)**
Awarded for securing the *first position* in Master of Technology
KIIT University, India

Chancellor's Merit Scholarship **(2004 – 2010)**
Awarded for *academic excellence*, Bachelor's and Master's studies
KIIT University, India.

Language Proficiency

- Bengali (native), English (fluent), Hindi (fluent), Arabic (A1), and Norsk (A2/B1)

Citizenship & Residency

Bangladeshi (Citizen), Norway (Permanent Residency)

References

Dr. Michael Cheffena
(PhD Supervisor)
Professor, Faculty of Engineering,
NTNU, Gjøvik 2815, Norway
michael.cheffena@ntnu.no, (+47) 45226765

Dr. Alok Mishra
(Distinguished Professor)
Professor, Faculty of Engineering,
NTNU, Gjøvik 2815, Norway
alok.mishra@ntnu.no, (+47) 46665761